JALA Academy
Assignment 5

Class Variables in Python

Python solution reference • Code examples are organized by the assignment questions.

How to use this document

Save the code examples into a .py file and run with Python 3. Where a program asks for input, enter values when prompted. For file-handling exercises, create the named text file in the same folder first.

1–5. Class and instance variables

```
class Student:
    school = "JALA Academy"  # class variable

    def __init__(self, name):
        self.name = name    # instance variable

print("Via class:", Student.school)
s1 = Student("Anu")
s2 = Student("Ravi")
print("Via instance:", s1.school)

# Assignment through an instance creates/updates an instance attribute,
# it does not change the class variable.
s1.school = "Local Campus"
print("s1.school:", s1.school)
print("s2.school:", s2.school)
print("Student.school:", Student.school)

# Change the class variable through the class itself.
Student.school = "JALA Academy - Updated"
print("Class:", Student.school)
print("s2 sees updated class value:", s2.school)
print("s1 still has its own value:", s1.school)

print("Instance variable:", s1.name)
print("Class variable:", Student.school)
```